

ScheduleSync — Project Workflow Report

Project: Numaligarh Refinery Expansion (Unit 3 & Offsites) · Client: Oil India Limited

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Total WBS Activities	Planned Date Range	Overall Duration	Contract Scope
36	2026-02-01 to 2026-02-28	28 days	Numaligarh Refinery Expansion (Unit 3 & Offsites) (Oil India Limited)

Schedule Breakdown by Engineering Discipline

Distribution of work packages and date spans across active project engineering disciplines.

Discipline	Activities	% of Total	Discipline Start	Discipline Finish	Span (Days)	Status Distribution
PIPING	12	33.3%	2026-02-01	2026-02-15	15 d	6 done · 4 active · 0 planned
CIVIL	12	33.3%	2026-02-01	2026-02-22	22 d	6 done · 3 active · 2 planned
ELECTRICAL	12	33.3%	2026-02-02	2026-02-28	27 d	3 done · 3 active · 3 planned

Workflow Interpretation: This report presents the baseline construction workflow for **Numaligarh Refinery Expansion (Unit 3 & Offsites)**. The execution is partitioned into **3 disciplines**. Each following section details the sequential progression of tasks within that discipline, followed by a comprehensive chronological sequence of all trades to highlight site handovers, interface milestones, and potential trade bottlenecks.

1. PIPING Discipline Workflow (12 Activities)

Planned activities for **PIPING** ordered chronologically by scheduled start date. Dependencies indicate preceding activities that must finish before initiation.

Activity ID	Activity Name	Location	Planned Start	Planned Finish	Duration	Predecessor Activity
L6-PIP-101	Erect Line 24 (12in CS pipe)	Unit 3	2026-02-01	2026-02-05	5 d	— (Root / Milestone)
L6-PIP-102	Erect Line 31 (8in CS pipe)	Unit 3	2026-02-02	2026-02-06	5 d	— (Root / Milestone)
L6-PIP-105	Install gate valve GV-01	Unit 3	2026-02-03	2026-02-06	4 d	— (Root / Milestone)
L6-PIP-106	Erect spool SP-044 on Line 24	Unit 3	2026-02-04	2026-02-07	4 d	L6-PIP-101
L6-PIP-104	Fit-up spool SP-045 (12in CS)	Unit 3	2026-02-04	2026-02-06	3 d	— (Root / Milestone)
L6-PIP-107	Weld joint J-12 on Line 31	Unit 3	2026-02-05	2026-02-08	4 d	L6-PIP-102
L6-PIP-112	Insulation wrap Line 40	Unit 3	2026-02-05	2026-02-07	3 d	— (Root / Milestone)
L6-PIP-103	Hydrotest Line 24 manifold	Unit 3	2026-02-06	2026-02-08	3 d	L6-PIP-101
L6-PIP-108	Erect spool SP-046 on Line 31	Unit 3	2026-02-07	2026-02-14	8 d	L6-PIP-102
L6-PIP-109	Fit-up weld joint J-14	Unit 3	2026-02-08	2026-02-13	6 d	— (Root / Milestone)
L6-PIP-110	Bolt-up flange FL-08 on Line 24	Unit 3	2026-02-08	2026-02-14	7 d	L6-PIP-101
L6-PIP-111	Pressure gauge PG-102 installation	Unit 3	2026-02-09	2026-02-15	7 d	L6-PIP-110

2. CIVIL Discipline Workflow (12 Activities)

Planned activities for **CIVIL** ordered chronologically by scheduled start date. Dependencies indicate preceding activities that must finish before initiation.

Activity ID	Activity Name	Location	Planned Start	Planned Finish	Duration	Predecessor Activity
L6-CIV-201	Excavate footing F-12	Unit 3	2026-02-01	2026-02-04	4 d	— (Root / Milestone)
L6-CIV-205	Excavate trench TR-03	Unit 3	2026-02-02	2026-02-05	4 d	— (Root / Milestone)
L6-CIV-202	Lay PCC for F-12 footing	Unit 3	2026-02-04	2026-02-06	3 d	L6-CIV-201
L6-CIV-206	Lay blinding concrete TR-03	Unit 3	2026-02-05	2026-02-07	3 d	— (Root / Milestone)
L6-CIV-204	Backfill around footing F-12	Unit 3	2026-02-06	2026-02-08	3 d	L6-CIV-202
L6-CIV-203	Cast column C-7 (RCC)	Unit 3	2026-02-06	2026-02-09	4 d	L6-CIV-201
L6-CIV-207	Reinforcement bar tying column C-7	Unit 3	2026-02-07	2026-02-13	7 d	— (Root / Milestone)
L6-CIV-209	Excavate pump house pit	Unit 3	2026-02-07	2026-02-14	8 d	— (Root / Milestone)
L6-CIV-208	Formwork erection column C-7	Unit 3	2026-02-08	2026-02-14	7 d	— (Root / Milestone)
L6-CIV-210	PCC pour pump house base	Unit 3	2026-02-09	2026-02-15	7 d	— (Root / Milestone)
L6-CIV-211	Grout column bases C-7 to C-10	Unit 3	2026-02-15	2026-02-20	6 d	— (Root / Milestone)
L6-CIV-212	Stormwater drain construction	Unit 3	2026-02-16	2026-02-22	7 d	— (Root / Milestone)

3. ELECTRICAL Discipline Workflow (12 Activities)

Planned activities for **ELECTRICAL** ordered chronologically by scheduled start date. Dependencies indicate preceding activities that must finish before initiation.

Activity ID	Activity Name	Location	Planned Start	Planned Finish	Duration	Predecessor Activity
L6-ELE-301	Pull 11kV cable - Substation to MCC	Substation	2026-02-02	2026-02-06	5 d	— (Root / Milestone)
L6-ELE-302	Install cable tray in MCC building	Unit 3	2026-02-04	2026-02-08	5 d	L6-ELE-301
L6-ELE-304	Earthing conductor laying Substation	Substation	2026-02-05	2026-02-08	4 d	— (Root / Milestone)
L6-ELE-303	Terminate 11kV feeder cable at switchgear	Substation	2026-02-05	2026-02-07	3 d	— (Root / Milestone)
L6-ELE-308	Install lighting fixtures Substation	Substation	2026-02-05	2026-02-07	3 d	— (Root / Milestone)
L6-ELE-309	Terminate motor cable P-101A	Substation	2026-02-05	2026-02-07	3 d	— (Root / Milestone)
L6-ELE-305	MCC panel positioning ground floor	Unit 3	2026-02-07	2026-02-14	8 d	— (Root / Milestone)
L6-ELE-306	Control cable termination panel A	Substation	2026-02-08	2026-02-15	8 d	— (Root / Milestone)
L6-ELE-307	Lighting distribution board LDB-1 install	Unit 3	2026-02-08	2026-02-14	7 d	— (Root / Milestone)
L6-ELE-310	Testing & commissioning MCC-A	Unit 3	2026-02-16	2026-02-24	9 d	— (Root / Milestone)
L6-ELE-311	Pre-commissioning substation lighting	Substation	2026-02-17	2026-02-23	7 d	— (Root / Milestone)
L6-ELE-312	Final punchlist closeout Electrical	Substation	2026-02-20	2026-02-28	9 d	— (Root / Milestone)

Cross-Discipline Master Execution Sequence

Integrated chronological timeline of all project trades. Displays interleaving of Civil, Piping, Electrical, and HSE activities over the project lifecycle to monitor critical trade handoffs.

Activity ID	Discipline	Activity Name	Location	Planned Start	Planned Finish	Duration	Predecessor
L6-PIP-101	PIPING	Erect Line 24 (12in CS pipe)	Unit 3	2026-02-01	2026-02-05	5 d	—
L6-CIV-201	CIVIL	Excavate footing F-12	Unit 3	2026-02-01	2026-02-04	4 d	—
L6-PIP-102	PIPING	Erect Line 31 (8in CS pipe)	Unit 3	2026-02-02	2026-02-06	5 d	—
L6-CIV-205	CIVIL	Excavate trench TR-03	Unit 3	2026-02-02	2026-02-05	4 d	—
L6-ELE-301	ELECTRICAL	Pull 11kV cable - Substation to MCC	Substation	2026-02-02	2026-02-06	5 d	—
L6-PIP-105	PIPING	Install gate valve GV-01	Unit 3	2026-02-03	2026-02-06	4 d	—
L6-PIP-106	PIPING	Erect spool SP-044 on Line 24	Unit 3	2026-02-04	2026-02-07	4 d	L6-PIP-101
L6-CIV-202	CIVIL	Lay PCC for F-12 footing	Unit 3	2026-02-04	2026-02-06	3 d	L6-CIV-201
L6-ELE-302	ELECTRICAL	Install cable tray in MCC building	Unit 3	2026-02-04	2026-02-08	5 d	L6-ELE-301
L6-PIP-104	PIPING	Fit-up spool SP-045 (12in CS)	Unit 3	2026-02-04	2026-02-06	3 d	—
L6-PIP-107	PIPING	Weld joint J-12 on Line 31	Unit 3	2026-02-05	2026-02-08	4 d	L6-PIP-102
L6-CIV-206	CIVIL	Lay blinding concrete TR-03	Unit 3	2026-02-05	2026-02-07	3 d	—
L6-ELE-304	ELECTRICAL	Earthing conductor laying Substation	Substation	2026-02-05	2026-02-08	4 d	—
L6-ELE-303	ELECTRICAL	Terminate 11kV feeder cable at switchgear	Substation	2026-02-05	2026-02-07	3 d	—
L6-PIP-112	PIPING	Insulation wrap Line 40	Unit 3	2026-02-05	2026-02-07	3 d	—
L6-ELE-308	ELECTRICAL	Install lighting fixtures Substation	Substation	2026-02-05	2026-02-07	3 d	—
L6-ELE-309	ELECTRICAL	Terminate motor cable P-101A	Substation	2026-02-05	2026-02-07	3 d	—
L6-PIP-103	PIPING	Hydrotest Line 24 manifold	Unit 3	2026-02-06	2026-02-08	3 d	L6-PIP-101
L6-CIV-204	CIVIL	Backfill around footing F-12	Unit 3	2026-02-06	2026-02-08	3 d	L6-CIV-202
L6-CIV-203	CIVIL	Cast column C-7 (RCC)	Unit 3	2026-02-06	2026-02-09	4 d	L6-CIV-201
L6-PIP-108	PIPING	Erect spool SP-046 on Line 31	Unit 3	2026-02-07	2026-02-14	8 d	L6-PIP-102
L6-CIV-207	CIVIL	Reinforcement bar tying column C-7	Unit 3	2026-02-07	2026-02-13	7 d	—
L6-CIV-209	CIVIL	Excavate pump house pit	Unit 3	2026-02-07	2026-02-14	8 d	—
L6-ELE-305	ELECTRICAL	MCC panel positioning ground floor	Unit 3	2026-02-07	2026-02-14	8 d	—
L6-PIP-109	PIPING	Fit-up weld joint J-14	Unit 3	2026-02-08	2026-02-13	6 d	—
L6-PIP-110	PIPING	Bolt-up flange FL-08 on Line 24	Unit 3	2026-02-08	2026-02-14	7 d	L6-PIP-101
L6-CIV-208	CIVIL	Formwork erection column C-7	Unit 3	2026-02-08	2026-02-14	7 d	—

Activity ID	Discipline	Activity Name	Location	Planned Start	Planned Finish	Duration	Predecessor
L6-ELE-306	ELECTRICAL	Control cable termination panel A	Substation	2026-02-08	2026-02-15	8 d	—
L6-ELE-307	ELECTRICAL	Lighting distribution board LDB-1 install	Unit 3	2026-02-08	2026-02-14	7 d	—
L6-PIP-111	PIPING	Pressure gauge PG-102 installation	Unit 3	2026-02-09	2026-02-15	7 d	L6-PIP-110
L6-CIV-210	CIVIL	PCC pour pump house base	Unit 3	2026-02-09	2026-02-15	7 d	—
L6-CIV-211	CIVIL	Grout column bases C-7 to C-10	Unit 3	2026-02-15	2026-02-20	6 d	—
L6-CIV-212	CIVIL	Stormwater drain construction	Unit 3	2026-02-16	2026-02-22	7 d	—
L6-ELE-310	ELECTRICAL	Testing & commissioning MCC-A	Unit 3	2026-02-16	2026-02-24	9 d	—
L6-ELE-311	ELECTRICAL	Pre-commissioning substation lighting	Substation	2026-02-17	2026-02-23	7 d	—
L6-ELE-312	ELECTRICAL	Final punchlist closeout Electrical	Substation	2026-02-20	2026-02-28	9 d	—